

Propose, Validate, Commit

State-First Architectures for Reliable Mixed-Output LLM Systems

Project family: Clio, LLM-Chess, Semantic Video Studio, ProveCalc, RPG-MCP, ChatDB, IndexFoundry,
Trace MCP, Synch MCP

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Date: 2026-05-16

Status: White paper v1.2

**Core claim: the failure is not that a model hallucinates; the failure is letting
hallucination commit.**

Abstract

Large language models are powerful generators but unreliable authorities. Modern model output is also no longer a single prose answer. A single run may contain narration, citations, tool calls, executable commands, JSON, visual stage directions, chess moves, equations, proof steps, benchmark records, captions, and media instructions. Treating that mixed stream as one trusted blob creates the core safety and reliability failure: invalid content can leak into persistent state, public display, or downstream artifacts.

The Mnehmos architecture relocates authority outside the model. The model proposes structured artifacts. External state preserves task context. Typed validators test candidate transitions. Deterministic reducers commit only accepted changes. Renderers and exporters display committed state rather than raw generation. Traces preserve the path from proposal to consequence.

The operating posture is trust, but verify. Models are trusted for generation, compression, translation, narration, and search. They are not trusted with direct authority over durable state, executable tools, public claims, or rendered media until independent controls verify the proposed transition.

The pattern appears across multiple implemented systems. Clio routes narration and Stagehand map commands into a sourced civic-intelligence renderer. LLM-Chess turns model moves, reasoning, commentary, Stockfish evaluation, and attack-channel behavior into games, benchmark data, and content artifacts. Semantic Video Studio treats cinematic state as the primary artifact and rendered MP4s as derivatives. ProveCalc keeps engineering calculation authority in SymPy, Pint, and worksheet verification gates. RPG-MCP lets the model narrate while a deterministic rules engine owns mechanics. ChatDB, IndexFoundry, Trace MCP, Synch MCP, and the multi-agent framework supply the surrounding substrate: memory, retrieval, schema tracing, orchestration, and proof-oriented persistence.

The practical result is a commit-boundary view of hallucination. Unsupported output may still be generated, but it is denied authority over durable state, public media, and exported records.

1. Executive Thesis

Reliable LLM systems should be designed around the transition from proposal to committed state, not around the apparent eloquence of the generated response.

- The model proposes candidates: text, commands, claims, edits, moves, assets, explanations, or plans.
- The state substrate remembers the world, task, sources, entities, files, moves, equations, or scene graph.
- Validators decide whether candidate transitions satisfy domain constraints.
- Reducers commit only accepted transitions.

- Renderers display committed state as UI, video, narration, charts, source cards, captions, or export files.
- Traces preserve rejected and accepted paths for debugging, replay, benchmark analysis, and governance.

LLM proposes
State remembers
Validator constrains
Reducer commits
Renderer performs
Trace preserves

This is not a claim that LLMs stop producing unsupported material. It is a claim that unsupported material should not receive commit authority. A system can tolerate noisy generation when state transitions are typed, checked, and reversible.

2. AI Development Context

The architecture aligns with several visible directions in the AI platform layer: agent runtimes, tool protocols, world models, and content provenance.

Industry direction	Relevant movement	Architectural consequence
Agent runtimes	Agents plan, call tools, collaborate across specialist components, and keep enough state for multi-step work.	The application must own orchestration, state, approvals, guardrails, and observability.
Tool protocols	MCP standardizes connections between AI applications and external systems such as files, databases, tools, and workflows.	Tool access needs typed boundaries so external power does not become unbounded model authority.
World models	Interactive world models can generate navigable environments from prompts, shifting attention from static answers to simulated state.	Black-box simulation creates value, but inspectable state is still required for audit, editing, and reuse.
Media provenance	C2PA and related standards address the source and history of media artifacts.	Provenance should be stored not only at final export, but throughout the production graph.

References for the industry context appear at the end. The important architectural pressure is consistent: serious AI products are becoming stateful, tool-using, multimodal, and media-producing. That increases the need for typed commitment boundaries.

3. The Mixed-Output Problem

The classic chatbot abstraction assumes the model returns a string. Current systems return mixed output. Some pieces are display text. Some are candidate state transitions. Some are citations. Some are tool arguments. Some are visual commands. Some are hidden or visible reasoning. Some are

useful only as benchmark telemetry. A safe architecture must separate those pieces before they reach sinks with consequences.

Output class	Example	Allowed sink	Required gate
Narration	A paragraph explaining a geopolitical event or chess move	Caption, transcript, TTS, article body	Claim support and style policy when factual
Executable command	map.highlight, object.scale, move.forward, database mutation	Runtime reducer or renderer	Schema, entity resolution, permission, and state validation
Claim	A factual statement about a source, battle, route, law, or metric	Claim graph, source card, public briefing	Source binding, uncertainty label, freshness check
Game move	e4, Nf3, Qxb7+	Chess reducer and event log	Legality check and board-state transition
Equation/edit	Add beam stress equation, solve for sigma	Worksheet graph	Unit consistency, constraint satisfaction, numeric residual
Asset instruction	Use rover_v003, import GLB, rotate camera	Scene graph or asset registry	Asset interface, hash, normalization, dry-run render gate
Benchmark telemetry	Attempts, centipawn loss, legal rate, latency, tokens	Dataset and leaderboard	Event mapping and aggregate computation
Rejected material	Invalid move, stale patch, unknown entity, unsupported citation	Private trace or error report	No public commit

The central reliability metric becomes mixed-output escape rate: how often invalid or unsupported model output reaches a public, persistent, or executable sink. The desired value is not zero bad generation. The desired value is zero bad commits.

Trust-but-verify turns mixed output into routed output. Each fragment is assigned to a sink only after the appropriate control tier has been applied: prose may need claim support, commands need schemas and permissions, assets need import gates, and game moves need legality checks.

4. Propose/Validate/Commit Architecture

The core loop is domain-independent:

- User intent
 - > model proposal
 - > typed intermediate representation
 - > validation against external state
 - > commit or rejection
 - > rendered/public output
 - > durable trace

4.1 Roles

Role	Function	Failure if absent
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Generator	Produces candidate structures, explanations, commands, edits, or moves.	No creativity or semantic compression.
State substrate	Stores durable task/world truth: sources, entities, board state, worksheet nodes, assets, scene graph, proof obligations.	The model must remember everything inside context and eventually drifts.
Validator	Tests proposals against schema, rules, sources, unit systems, game legality, import gates, or proof obligations.	Invalid output can become real.
Reducer	Applies valid transitions deterministically.	The same proposal may produce inconsistent consequences.
Renderer/exporter	Turns committed state into UI, video, narration, PGN, CSV, JSON, PDF, captions, or source cards.	Display can diverge from truth state.
Trace	Records accepted and rejected transitions with enough detail for replay and evaluation.	Failures become anecdotes instead of data.

4.2 Constraint as Architecture

A prompt asks the model to behave. A schema defines the shape of permissible output. A validator enforces the shape against actual state. A reducer makes only valid transitions durable. An audit trail preserves the consequence. Reliable systems depend on the latter four layers, not on the hope that the prompt will be obeyed forever.

4.3 Trust, But Verify

Trust is assigned by role, not by reputation. A model can be trusted as a generator without being trusted as a committer. It can draft a claim, suggest a move, propose an equation, narrate a scene, or request an asset. Verification happens at the boundary where that proposal would become persistent, executable, public, or expensive.

The practical rule is simple: trust the model to propose useful candidates; verify before those candidates touch a sink. In Clio, that sink is the public show event stream. In LLM-Chess, it is the board reducer, benchmark store, and export package. In SVS, it is the four-plane state pack, render manifest, and asset registry. In ProveCalc, it is the verified worksheet graph. In RPG-MCP, it is canonical world state.

This framing avoids two common mistakes. Blind trust lets plausible prose mutate reality. Blanket distrust wastes the model as a semantic engine. Trust-but-verify keeps the useful part: fast proposal generation, followed by independent validation before commitment.

4.4 Hierarchy of Controls for LLM Systems

Safety-critical industries use a hierarchy of controls: eliminate a hazard where possible, substitute safer mechanisms, add engineering controls, then rely on administrative process and last-mile warnings only after stronger controls have been exhausted. Reliable LLM systems should follow the same order. The safest control is not a better disclaimer; it is removing the model path that can directly cause the bad outcome.

Control tier	LLM-system interpretation	Mnehmos pattern
Elimination	Remove direct model authority over dangerous or durable sinks.	Raw model text cannot directly mutate the database, board, worksheet, world state, asset registry, or public render stream.
Substitution	Replace free-form authority with typed intermediate representations.	Stagehand commands, chess moves, semantic patches, worksheet nodes, action enums, source records, and state packs.
Engineering controls	Enforce constraints mechanically at the boundary.	Schemas, validators, reducers, import gates, source/entity registries, unit gates, legality checks, Stockfish evaluation, manifest hashes.
Administrative controls	Use review workflows, release gates, documentation, runbooks, and human approval where judgment is required.	Citation review, asset promotion, scenario approval, benchmark profile selection, versioned export/release packages.
Last-mile warnings	Labels, confidence displays, UI warnings, and disclaimers.	Useful for interpretation, but not allowed to substitute for missing validation gates.

The order matters. A warning label on unverified output is weaker than a validator that prevents that output from reaching the public stream. A human review checklist is weaker than a schema that makes invalid structure unrepresentable. The design goal is to push controls upward in the hierarchy wherever the domain allows it.

5. Case Study: Clio

Clio is a rendered civic-intelligence system: a typed world canvas where sourced claims, entities, geography, narration, and visual operations become public map briefings. It belongs to the same family as a newsroom graphics desk, a GIS dashboard, a live explainer, and an agentic research workstation.

5.1 Stagehand as a Trust Transducer

Stagehand is the key mechanism. Narration and visual directions travel together, but they are not trusted equally. Human-readable narration can become captions or TTS only after factual treatment rules apply. Map effects become public only after the command parser and validator confirm action schema, argument shape, entity references, source references, and visual treatment rules.

Narration stream with inline Stagehand command

- > parser separates prose from command
- > command schema validates action and arguments
- > entity/source registry resolves references
- > public show event is emitted
- > renderer performs only accepted effects

5.2 Public/Private Boundary

Clio separates private production from public display. Private space can contain raw model output, rejected commands, unresolved claims, repair attempts, source fetches, and validation failures. Public space receives only validated show events: captions, map effects, source cards, gap cards, safe failures, and exportable show packages.

5.3 Product Meaning

The product is not a chatbot answer. The product is a validated world-state update rendered as media. That makes Clio a natural endpoint for mixed-output routing: factual claims go to a claim graph, source references go to source cards, Stagehand commands go to the map runtime, narration goes to TTS/captions, and all accepted transitions go to trace.

6. Case Study: LLM-Chess

LLM-Chess is both an LLM benchmarker and a content generator. It runs LLM-vs-LLM or LLM-vs-engine chess games, records model behavior, evaluates moves, aggregates metrics, and produces replayable media artifacts. The board is the world state. Chess legality is the validator. Stockfish is an oracle. The event log is provenance. Commentary, board annotations, TTS, PGN, CSV, JSON, and ZIP exports are the rendering layer.

6.1 Benchmarking Layer

The benchmark system maps games into structured records. It tracks model identity, prompt level, output format, constraint mode, opponent type, attacks, fog-of-war settings, advisor behavior, moves, illegal attempts, FEN before/after, centipawn loss, legal-first-attempt rate, response time, token counts, and result. Aggregates compute Elo estimates, win rate, legal rate, centipawn loss, response-time metrics, confidence/assessment accuracy, sycophancy-style advisor deference, attack resilience, and related behavioral measures.

System element	LLM-Chess embodiment	Architectural role
State substrate	Board position, FEN, move history, game event log, tournament store, benchmark database.	Durable truth surface.
Validator	chess.js legality, game reducer, Stockfish evaluation, retry handling.	Rejects illegal moves and evaluates quality.
Mixed output	Move, reasoning, confidence, plan, threat notes, commentary, board annotations, attack metadata.	Model output is partitioned rather than trusted as one blob.
Benchmark sink	Game records, move records, aggregates, leaderboard, findings, reports.	Transforms play into research data.
Content sink	Live commentary, move batching, filler generation, optional TTS, replay, PGN/CSV/JSON/ZIP exports.	Transforms benchmark traces into public-facing media.

6.2 Content Generation Layer

LLM-Chess is close to Clio because both systems convert validated state into a show. Clio stages maps. LLM-Chess stages a board. Clio narrates geopolitical movement. LLM-Chess narrates chess movement. Clio needs source cards; LLM-Chess needs board truth and engine evaluation. Both can stream live, replay later, and export durable artifacts.

6.3 Adversarial Value

The chess environment is unusually valuable because wrongness is measurable. Illegal moves, poor centipawn decisions, overconfidence, sycophancy toward advisors, fog-of-war errors, prompt-injection susceptibility, FEN corruption, and context-bloat failure can all be recorded. LLM-Chess therefore supplies a controlled arena for the same reliability question Clio faces in civic media: can the system preserve useful model output while preventing bad output from becoming authoritative?

7. Case Study: Semantic Video Studio

Semantic Video Studio extends the same architecture into cinematic generation. End-to-end video models treat the MP4 as the unit of work. SVS treats structured cinematic state as the unit of work and the MP4 as a derived artifact.

7.1 Four-Plane State

Plane	File	Role
Scene graph	scene_graph.json	Object IDs, transforms, cameras, lights, environment.
Asset manifest	asset_manifest.json	Reusable typed assets with interfaces, capabilities, and anchors.
Timeline	timeline.json	Typed beats and actions over object, camera, and light IDs.
Render plan	render_plan.json	Engine, resolution, FPS, samples, output path, and render settings.

A render is a function of those planes plus the renderer code and schemas. Semantic edits become typed patches with base hashes. External assets pass through an import/normalization/validation/preview/registration gate. Provenance manifests hash every input and output. The rendered video becomes reproducible, partially regenerable, and auditable.

8. Supporting Systems

System	What the model proposes	What validates or commits	Reliability lesson
ProveCalc	Equations, worksheet edits, assumptions, solve goals, explanations.	SymPy, Pint, unit gates, constraint checks, numeric residuals, worksheet history.	The LLM can suggest engineering structure, but deterministic math owns correctness.
RPG-MCP	Narration, intended	Action router, schemas,	Imagination remains

	actions, spell usage, combat choices, inventory/world interactions.	deterministic D&D rules engine, dice, HP, initiative, inventory, quests.	creative while mechanics remain enforceable.
ChatDB	Proof steps, branches, obligations, candidate reasoning traces.	Persistent proof database, verification modules, accepted/rejected records, obligation tracking.	Proof search becomes a stateful workbench rather than a disposable conversation.
IndexFoundry	Requests to connect, extract, normalize, index, and serve knowledge.	Deterministic artifact hashes, pipeline manifests, generated MCP servers, Docker/Railway packaging.	RAG becomes a reproducible build artifact rather than a vague retrieval trick.
Trace MCP	Schema and API assumptions across services.	Static tracing across producers and consumers, OpenAPI/tRPC/MCP/Zod/Express/FastAPI/etc.	Typed mismatch detection moves reliability earlier in the software lifecycle.
Synch MCP	Memory writes, handoffs, bug records, coordination state.	SQLite/Markdown-backed persistent memory, search, file locks, bug tracking, handoff records.	Agents become more reliable when project memory exists outside the chat context.
Multi-Agent Framework	Plans and specialist outputs across central, somatic, autonomic, and reflex layers.	Layered orchestration, memory, tool interfaces, reflex validation, templates.	Agent systems need nervous-system structure: cognition, body, background memory, and reflex gates.

9. Evidence Matrix

The systems differ in maturity and validator strength. The shared pattern is architectural rather than a claim that every domain has the same formal rigor.

Domain	Committed state	Validator strength	Public/rendered output	Primary metric
Civic media / Clio	Entities, source cards, Stagehand effects, public show events.	Schema/entity/source/runtime validation; claim auditor remains a key next gate.	Map briefing, captions, source panels, export packages.	Invalid public effect escape rate; claim-source binding rate.
Chess / LLM-Chess	Board state, moves, game log, benchmark records.	Very strong legality gate; strong Stockfish quality oracle.	Live board, commentary, TTS, PGN/CSV/JSON/ZIP, leaderboard.	Legal rate, Elo, centipawn loss, attack resilience, response time.
Cinematic generation / SVS	Four-plane state pack, assets, timeline, render plan, provenance manifest.	Schema validation, cross-file invariants, import gate, basics gate.	MP4, captions, manifests, previews.	Partial regen correctness; state-to-render reproducibility.
Engineering / ProveCalc	Worksheet graph and verified calculation nodes.	Strong symbolic/unit/numeric gates.	Engineering worksheet, exports, audit trail.	Unit consistency, constraint satisfaction, residual checks.
RPG simulation / RPG-MCP	World state, combat state, inventory, quests, NPC memory.	Strong game-rule validator for mechanics.	Narrated RPG session and tool results.	Rule legality, persistence, mechanical consequence.
Proof / ChatDB	Proof state,	Variable:	Proof workspace and	Closed obligations,

	obligations, accepted/rejected steps.	symbolic/formal validators depending on configured proof layer.	reasoning dataset.	verified steps, rejection trace.
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10. Evaluation Program

State-first systems should be evaluated at the commit boundary. The following measures generalize across the Mnehmos project family.

Metric	Definition	Where it applies
Mixed-output escape rate	Invalid or unsupported model output that reaches a public, persistent, or executable sink.	Clio, LLM-Chess, RPG-MCP, ProveCalc, SVS.
Commit rejection precision	Rejected proposals that are truly invalid or unsafe.	All validator-backed systems.
Commit rejection recall	Invalid proposals that validators successfully catch.	All validator-backed systems.
Source binding rate	Public factual claims with explicit source support and uncertainty labels.	Clio, research tools, civic media.
State reproducibility	Same state pack or worksheet produces the same output within deterministic bounds.	SVS, ProveCalc, IndexFoundry.
Partial regeneration accuracy	Editing one state plane invalidates only the correct downstream artifacts.	SVS, Clio show packages, software build pipelines.
Oracle disagreement capture	Cases where model explanation conflicts with a stronger oracle.	LLM-Chess, ProveCalc, proof systems.
Human review compression	Amount of human effort moved from generation to review of typed diffs, traces, and rejected proposals.	All systems.
Control-tier coverage	The strongest hierarchy-of-controls layer guarding each public, persistent, or executable sink.	All systems; especially release gates, renderers, databases, and tool APIs.

11. Why LLM-Chess Belongs Beside Clio

Clio and LLM-Chess occupy the same structural niche from different directions. Clio starts from civic explanation and becomes a source-grounded show engine. LLM-Chess starts from a constrained game and becomes a benchmark/content engine. Both transform model output into public media only after state has been validated.

Dimension	Clio	LLM-Chess
Truth surface	World/entity/source state.	Board/FEN/game state.
Validator	Stagehand schemas, entity/source registries, visual treatment rules.	Chess legality, reducer, Stockfish oracle, retry handling.
Mixed output	Narration, source claims, map commands, visual effects, captions.	Moves, reasoning, confidence, plans, commentary, annotations, metrics.
Public artifact	Briefing, source cards, globe/map	Live game, commentary/TTS, replay,

	show, export package.	PGN/CSV/JSON/ZIP, leaderboard.
Research value	Civic-trust evaluation under source and claim pressure.	Model behavior evaluation under legal, strategic, adversarial, and oracle pressure.

The chess domain gives unusually clean labels: legal or illegal, better or worse by engine evaluation, confident or overconfident, advice-following or independent, attacked or clean. Those labels make LLM-Chess a practical benchmark generator. The content layer makes the same traces useful as media. That dual identity is exactly the Clio pattern: reliable public output is a rendering of validated state.

12. Limitations

- The architecture blocks invalid commits; it does not prevent invalid thoughts or unsupported proposals from being generated.
- Validator quality determines system quality. A weak validator becomes a weak boundary.
- Schemas can encode mistaken assumptions. State-first architecture does not eliminate governance or domain review.
- Clio still needs a complete claim-level citation gate, durable trace release gate, and export pipeline to match its final civic-media ambition.
- SVS trades photorealism for editability, provenance, and partial regeneration.
- LLM-Chess is a powerful behavioral benchmark, but chess is still a bounded domain; civic reasoning has messier truth conditions.
- External sources, assets, and memories can be poisoned. Provenance and quarantine are necessary, not optional.
- Human judgment remains part of the system, especially for factual framing, ethical release, aesthetic quality, and institutional accountability.

Trust-but-verify does not mean every domain can be fully automated. Some claims, assets, and public interpretations still require administrative controls and human review because no available validator captures the whole risk surface.

The hierarchy of controls is a design target, not a guarantee. Moving a risk upward from warning to schema, from checklist to validator, or from human memory to durable provenance is progress only when the new control actually covers the relevant failure mode.

13. Conclusion

The next reliable AI products will not be built by asking models to become perfect authorities. They will be built by placing models inside systems that remember, validate, commit, render, and preserve traces.

The right trust posture is neither blind acceptance nor blanket refusal. It is trust, but verify, implemented as a hierarchy of controls: eliminate direct commit authority where possible, substitute typed proposals for free-form power, enforce boundaries mechanically, document and review what remains, and reserve warnings for residual risk rather than primary safety.

Clio shows the civic-media form of that pattern. LLM-Chess shows the benchmark-and-content form. Semantic Video Studio shows the state-first cinematic form. ProveCalc shows the engineering-calculation form. RPG-MCP shows the rules-grounded simulation form. ChatDB, IndexFoundry, Trace MCP, Synch MCP, and the multi-agent framework supply the surrounding infrastructure for memory, retrieval, verification, and orchestration.

The model may propose anything.
Only validated state becomes real.

Appendix A: Repository Evidence Notes

Implementation files and project knowledge documents anchor the system descriptions. Public repositories are used where available; milestone summaries, architecture notes, and project knowledge records are valid evidence for active or private systems.

System	Evidence source	Observed capability
LLM-Chess	README.md; src/engine/events.ts; src/benchmark/mapper.ts; src/benchmark/aggregates.ts; src/commentary/commentaryQueue.ts; src/utis/export.ts	Tournament gauntlet, multi-provider LLM support, benchmark metrics, attack channels, event logs, commentary, TTS, and exports.
Clio	README.md; docs/reference/architecture.md; docs/reference/stagehand.md; src/stagehand/validator.ts; src/production/types.ts; src/production/renderPlan.ts	Rendered typed world model, Stagehand protocol, production/public event boundary, command validation, public show events.
ProveCalc	README.md; src-tauri/src/verification/mod.rs	Worksheet verification, unit checks, constraint checks, numeric residuals, deterministic compute authority.
RPG-MCP	README.md; src/utis/action-router.ts; src/server/consolidated/combat-manage.ts	Rules-enforced RPG backend, consolidated action routing, combat/inventory/world-state validation.
ChatDB	README.md; src-tauri/src/db/schema.rs; src-tauri/src/verification/mod.rs	Persistent proof workspace, database schema, verification module, accepted/rejected reasoning state.
IndexFoundry MCP	README.md	Deterministic vector index/RAG pipeline with content-hashed artifacts and deployable MCP server output.
Trace MCP	README.md	Schema mismatch tracing across OpenAPI, tRPC, MCP/Zod, Express, FastAPI, GraphQL, gRPC, and related interfaces.
Synch MCP	README.md	Persistent memory, project coordination, search, file locks, bug

		tracking, and handoff records.
Multi-Agent Framework	README.md	Central/somatic/autonomic/reflex architecture for multi-agent orchestration and validation.
Semantic Video Studio	Project knowledge docs: SVS M0-M11 milestone summary; basics gate notes; schema inventory; Blender/render pipeline notes; asset registry/import gate descriptions.	State-first cinematic pipeline, four-plane JSON state pack, semantic patches, asset backlot, basics gate, provenance manifests, and partial regeneration.

Appendix B: Industry References

1. OpenAI Agents SDK documentation: agents are applications that plan, call tools, collaborate across specialists, and keep enough state for multi-step work; SDK path is suited when the application owns orchestration, tool execution, approvals, and state.
<https://developers.openai.com/api/docs/guides/agents>
2. Model Context Protocol introduction: MCP is an open-source standard for connecting AI applications to external systems such as files, databases, tools, and workflows.
<https://modelcontextprotocol.io/docs/getting-started/intro>
3. Google DeepMind Genie 3 announcement: Genie 3 is described as a general-purpose world model that generates interactive environments navigable in real time at 24 fps and 720p for a few minutes. <https://deepmind.google/blog/genie-3-a-new-frontier-for-world-models/>
4. C2PA specifications: C2PA develops technical standards for certifying the source and history, or provenance, of media content. <https://spec.c2pa.org/specifications/specifications/2.2/index.html>